

ACIRP

Autonomous Civic Incident Platform

Ensuring safe streets through agentic civic action.

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Integrations: Gemini 2.5, Firebase, Google Cloud, Mem0, Keploy

THE CIVIC CHALLENGE

"Urban India records millions of civic complaints annually, yet delayed routing and lack of verification leave many hazards unresolved."

- Municipal departments are overwhelmed by duplicate, un-prioritized reports.
- Database outages or portal timeouts disconnect citizens from municipal cells.
- Lack of automated validation makes verification of field resolution expensive.

HOW ACIRP HELPS

- Perception Agent: Structuring photos into prioritized complaints.
- Generates structured complaint reports and escalation petitions with human-in-the-loop review.
- Resilience Engine: Gracefully escalates to safety dispatchers on server outage.
- Resolution Proof: Automates audit verification via visual comparison.

System Workflow & Architecture

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WHY AGENTIC AI?

* Vision-Based Recognition:

Structures raw incident photos into categorised details automatically.

* Autonomous Context Routing:

Dynamically routes files to target helplines based on location/type.

* Resolution Verification:

Analyzes before/after visual proof to ensure roads/bins are clear.

CORE TECHNOLOGY STACK

AI Models & SDK:

Gemini 2.5 Flash, Google GenAI SDK (Google AI Studio)

Primary Database:

Real-time incident database (Google Cloud Firestore)

Asset Hosting:

Firebase Storage (visual proofs) and Firebase Hosting CDN

Mem0 (Agent Memory Integration)

- Remembers complainant profiles across sessions.
- Identifies local ward hazard repeating histories.
- Real-time 'Memory Context' timelines generated automatically.

Keploy (API Test Mocking)

- Records real API calls and generates mocks automatically.
- Tracks custom headers (x-competition) for route routing.
- Ignores timestamps to prevent fake testing regression checks.

Market Impact & Operational Value

- Helps municipal officials, currently drowning in duplicate complaints, organize their work systematically.
- Improves transparency through automated tracking and visual verification.
- Allows ward representatives to view structured reports with human-in-the-loop approval.
- Reduces auditing costs and manual field inspection cycles.
- Highly scalable architecture: applicable to any municipality globally.